Problem-solving and decision-making

July, 2011

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Importance of good problem-solving and decision-making

- Get to answers faster
- Use time and resources more efficiently
- Have greater impact on the business
- Free up time to do other things

Objectives and approach for session

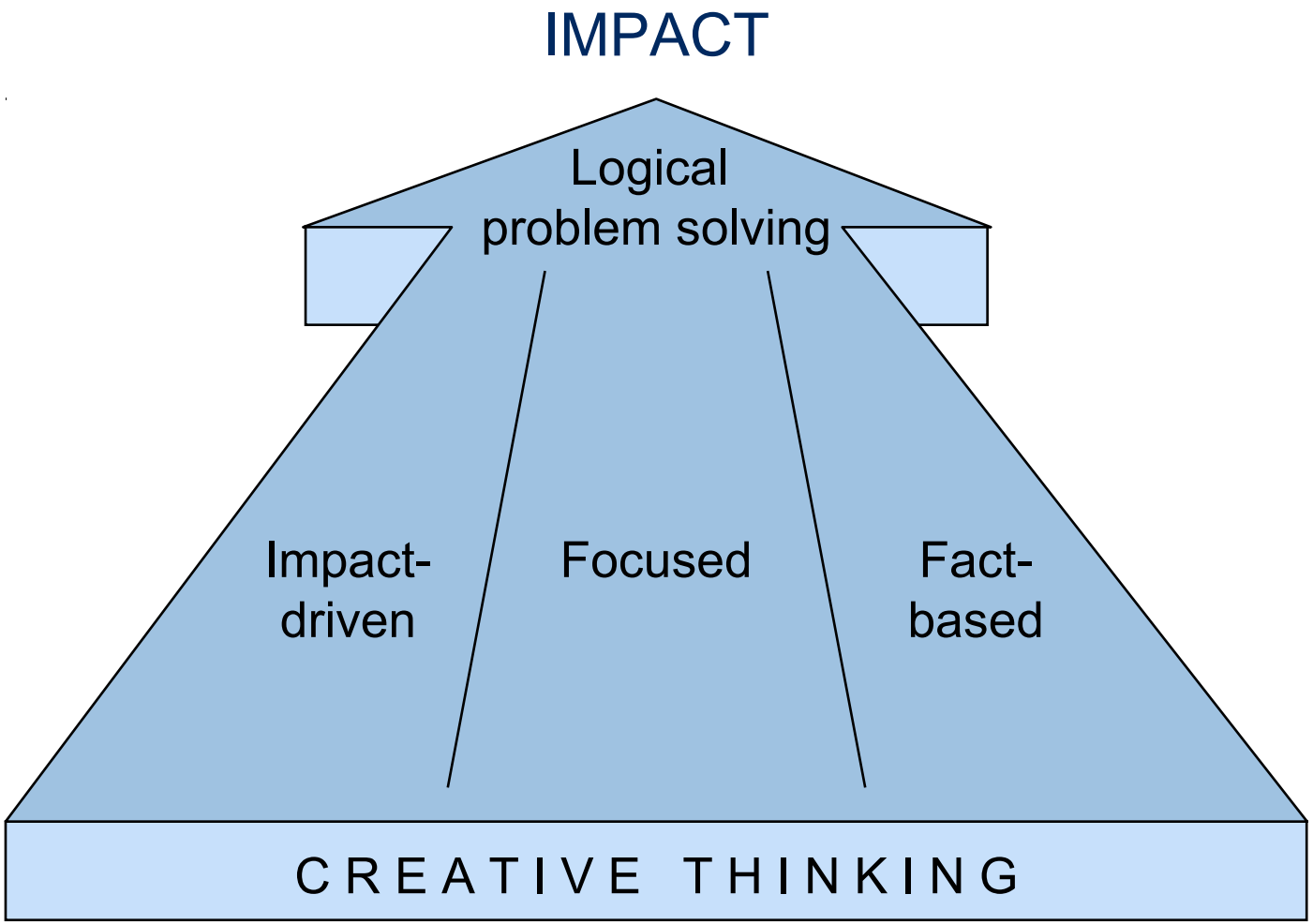
Objectives

- Introduce an approach/tools you can add to your toolkit of skills and use for many types of problem solving

Approach

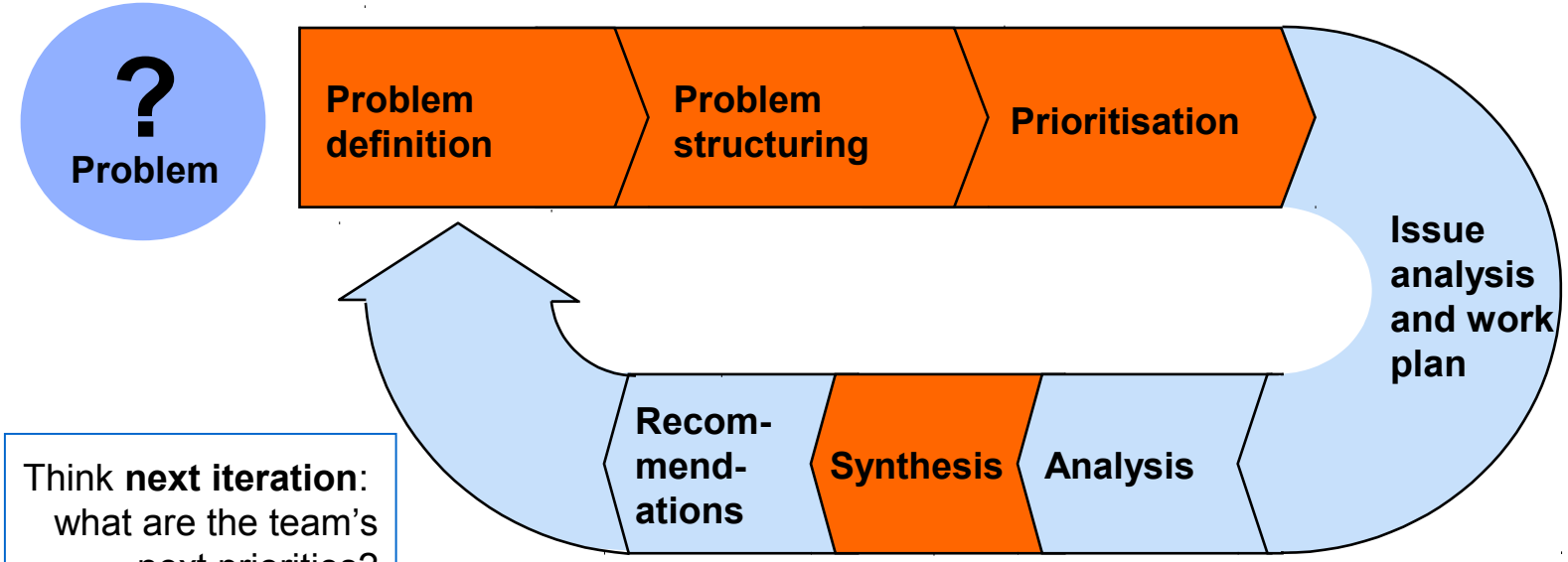
- Learn key elements of approach through plenary lecture and discussion

McKinsey's problem solving approach



The problem-solving cycle

Focus for today



Think **next iteration**: what are the team's next priorities?

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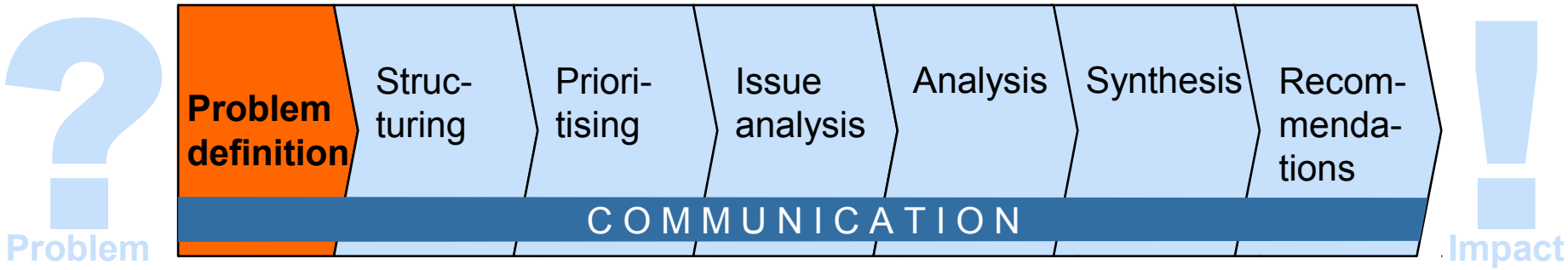
Step 1: Define the problem

Think **IMPACT**:
What is the question you are trying to answer?



What is a good problem definition?

A ‘problem definition’ is a way to frame a problem so that we are clear up front on what we are trying to solve, and what success will look like

- What are characteristics of a good problem definition?
- What are “must-have” elements to
  - Ensure we get to the best answer?
  - Ensure our answer is practical and implementable?
- What are the common pitfalls?

Problem statements help sharpen your thinking about the problem

What are we trying to achieve?

Focuses the work and ensures that findings can be acted on. The more specific the statement the better, provided that it is not so narrow that the “wrong” problem is addressed.

1 Background and context

Comments on the “situation” and “complication” facing the key decision-maker (e.g., industry trends, relative position in the industry).

2 What is success?

Refers to the basis on which the key decision-maker will decide whether to act on the recommendations (e.g.,financial returns, effect on staff, market share growth).

3 Stakeholders

Identifies primary decision makers (e.g., CEO, Division Manager) as well as internal and external parties who can affect implementation (e.g., shareholders)

4 Potential challenges

Indicates what will not be included in the project (e.g., international markets, R&D activities). Also defines the limits to the set of solutions that can be considered, including degree of accuracy of analysis (i.e, 80/20 rule vs. 90/10).

5 Where will we find information and help?

Describes implementation challenges (e.g., labour relations, communication practices, risk aversion) that must be addressed to change beliefs and behaviors

One idea of what a good problem definition looks like…

What are we trying to achieve?

The basic question we are trying to solve. It should be SMART: specific, measurable, action-oriented, relevant and time-bound.

1 Background and context

What is the current situation?
What has happened so far?
Why is this problem is being addressed now?

3 Stakeholders

Who are the decision makers?
Who needs to be involved?

2 What is success?

What does a good final product look like?
What are we trying to deliver?

4 Potential challenges

What are the potential issues or problems that we can forsee and plan to mitigate?

5 Where will we find information and help?

Where does expertise and knowledge, exist (Internal and external)? e.g., work that has already been done, internal experts who we should draw on, published reports and papers

Problem statement

Characteristics of good problem statement

- **S**pecific
- **M**easurable
- **A**ction-oriented
- **R**elevant (to the key problem)
- **T**ime-bound

Example problem statement: Oilco refinery

The Oilco refinery is suffering from poor profitability despite a strong market niche position

Statement of fact

Should the Oilco refinery improve its deteriorating position?

Not disputable

Can the Oilco refinery be managed differently to increase profitability?

Too general

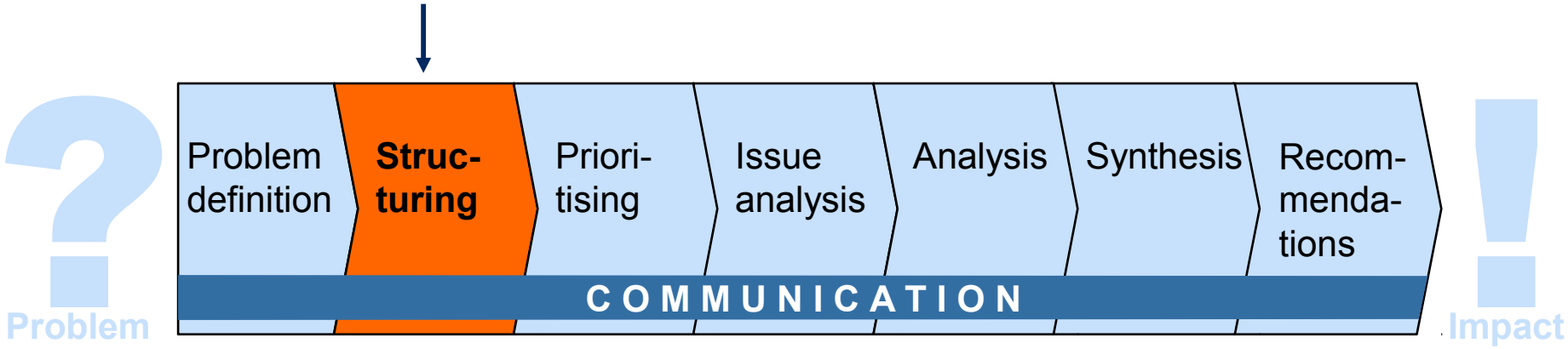
Good problem solving: Oilco refinery

What opportunities exist for Oilco to improve profitability by $40 million per year through overhead rationalisation, operational improvements, or restructuring non-core assets?

- **S**pecific
- **M**easurable
- **A**ction-oriented
- **R**elevant (to the key problem)
- **T**ime-bound

Step 2: Structure the problem

Think **DISAGGREGATION** and **HYPOTHESIS**:
What could the key elements of the problem be?



Problem solving requires we answer a number of difficult questions

What issues should we think about?

What steps will we need to take?

How will we build our smart cities?

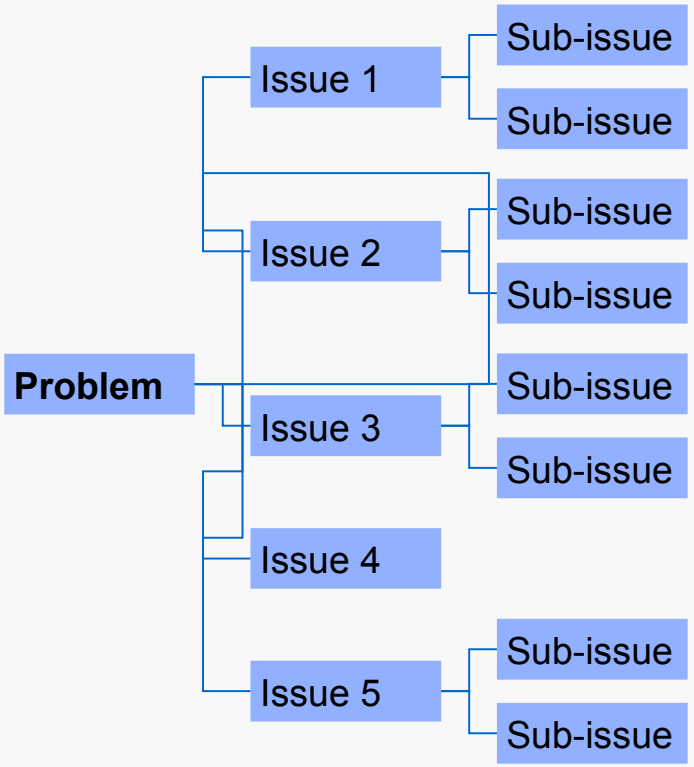
…?

What is an issue tree and why is it important?

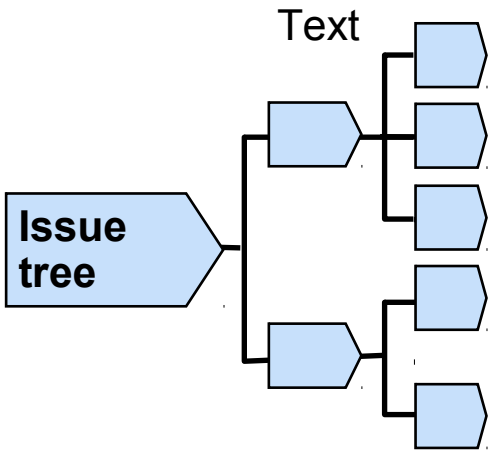
A problem solving tool that breaks a problem into discrete chunks

Why use logic trees?

- To break a problem into component parts
- To ensure integrity of the problem solving is maintained
- To build a common under-standing within the team of the problem solving framework
- To help focus team efforts

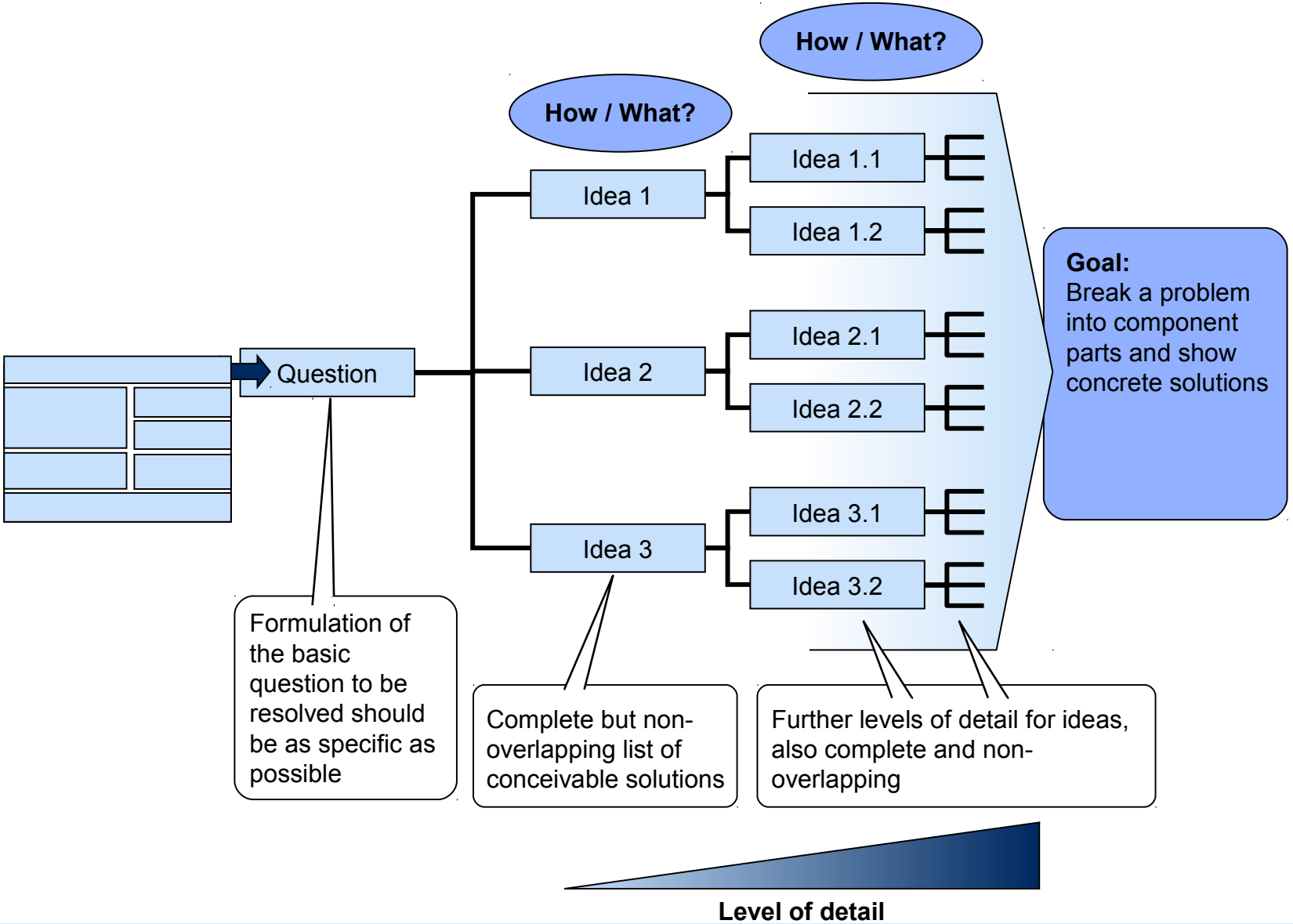


The issue tree is a valuable tool to disaggregate a problem into core sub-questions that need to be addressed to answer the main question



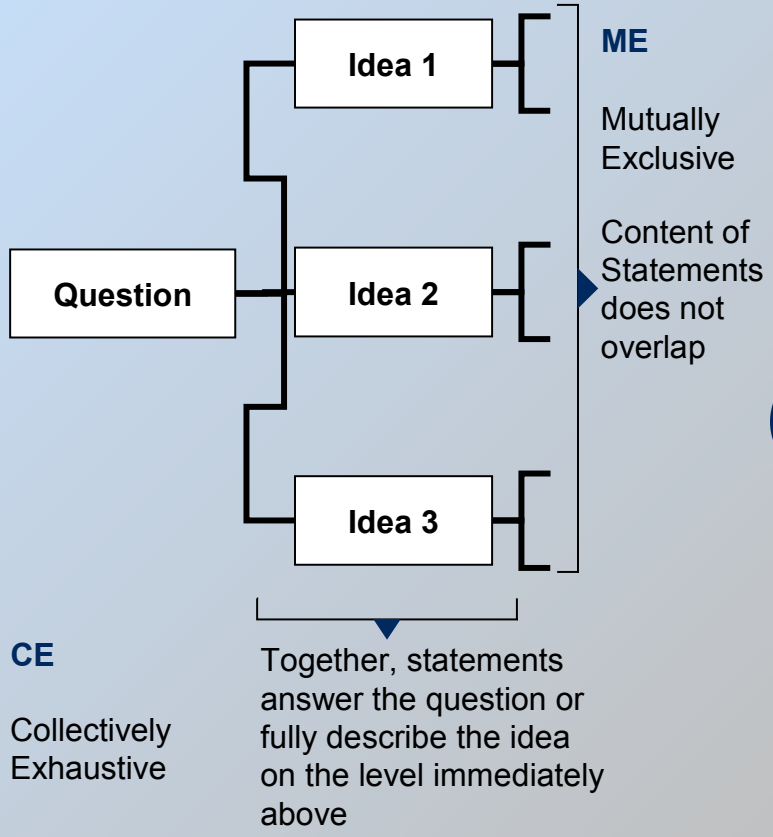
Description	Why use it?	When to use it?
▪ Decomposes an issue into smaller sub-issues **(e.g., measures, criteria)** ▪ Sub-issues answer the question "What?" or "How?"	▪ Helps disaggregate problem into individual pieces ▪ Helps divide the work among team members	▪ Early in the problem solving process, when you know little about the problem

Basic principle of the issue tree

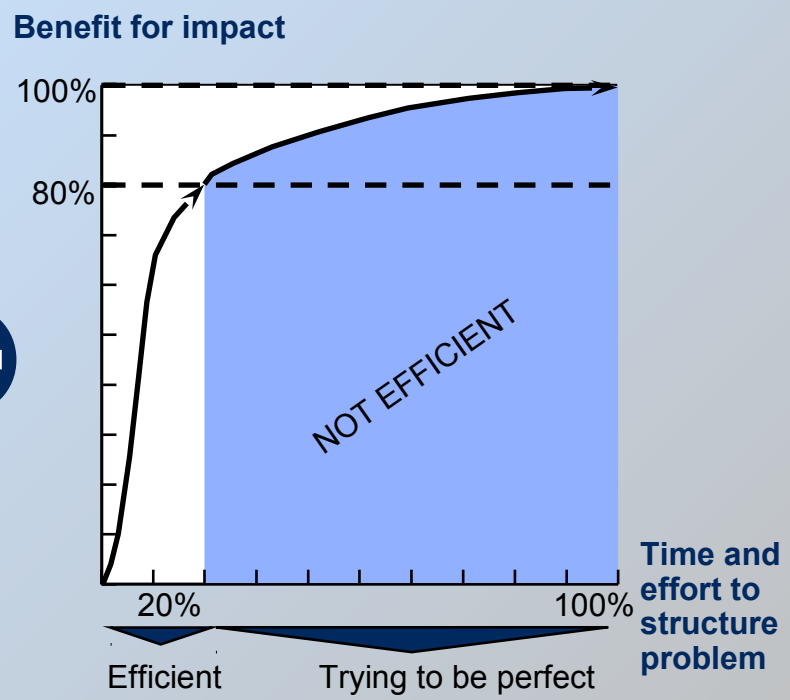


Good issue trees meet MECE and 80/20 rule requirements

Mutually Exclusive Collectively Exhaustive



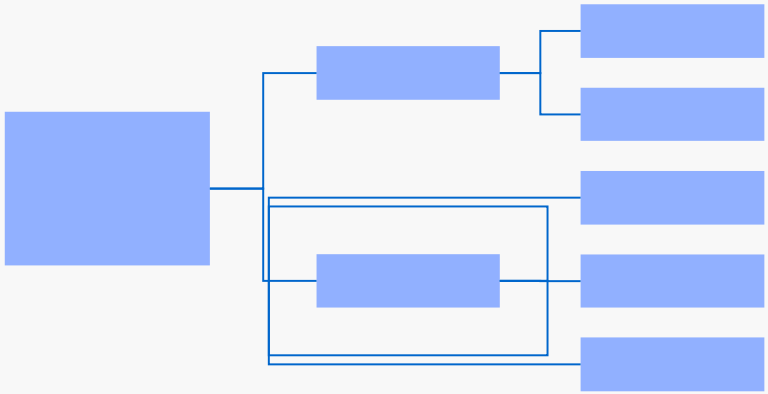
80/20 rule



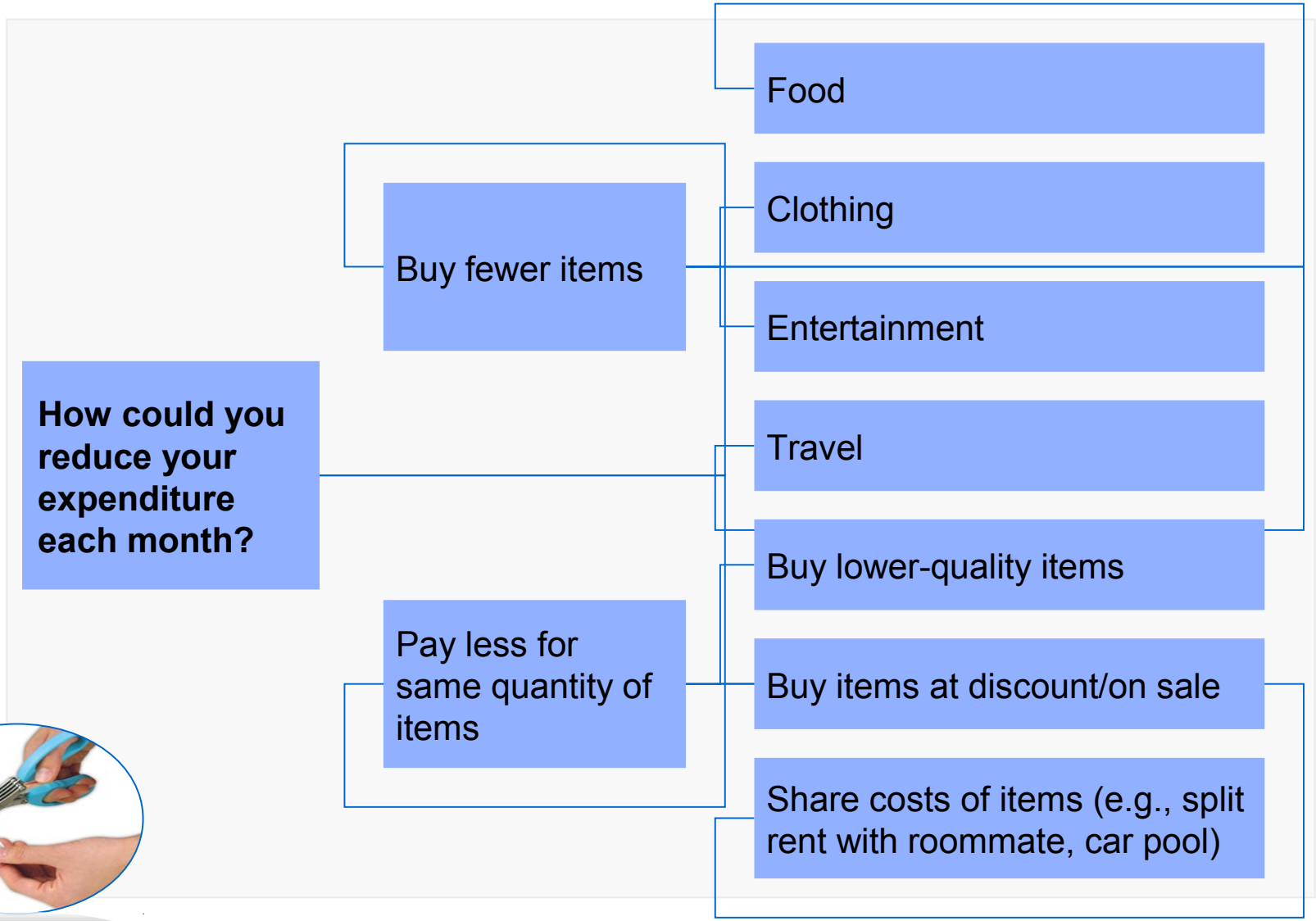
Let’s try out an example

Create an issue tree to address the following:

How can you reduce monthly shopping expenses?

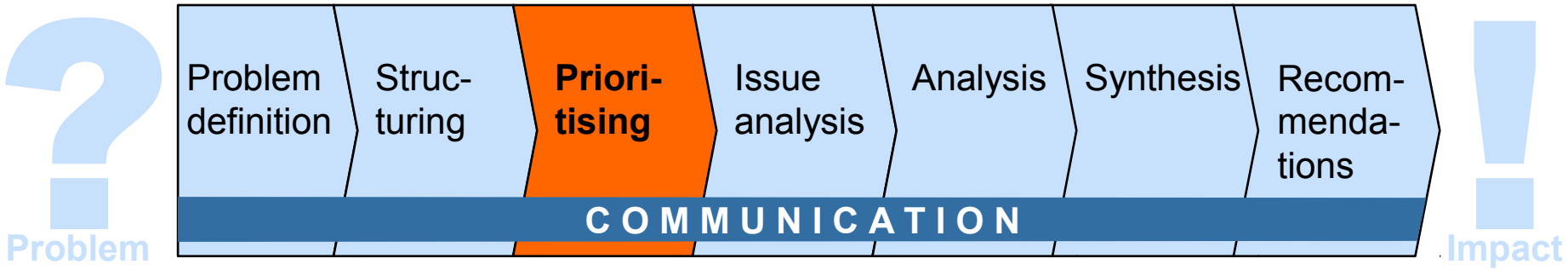


Example Issue tree solution

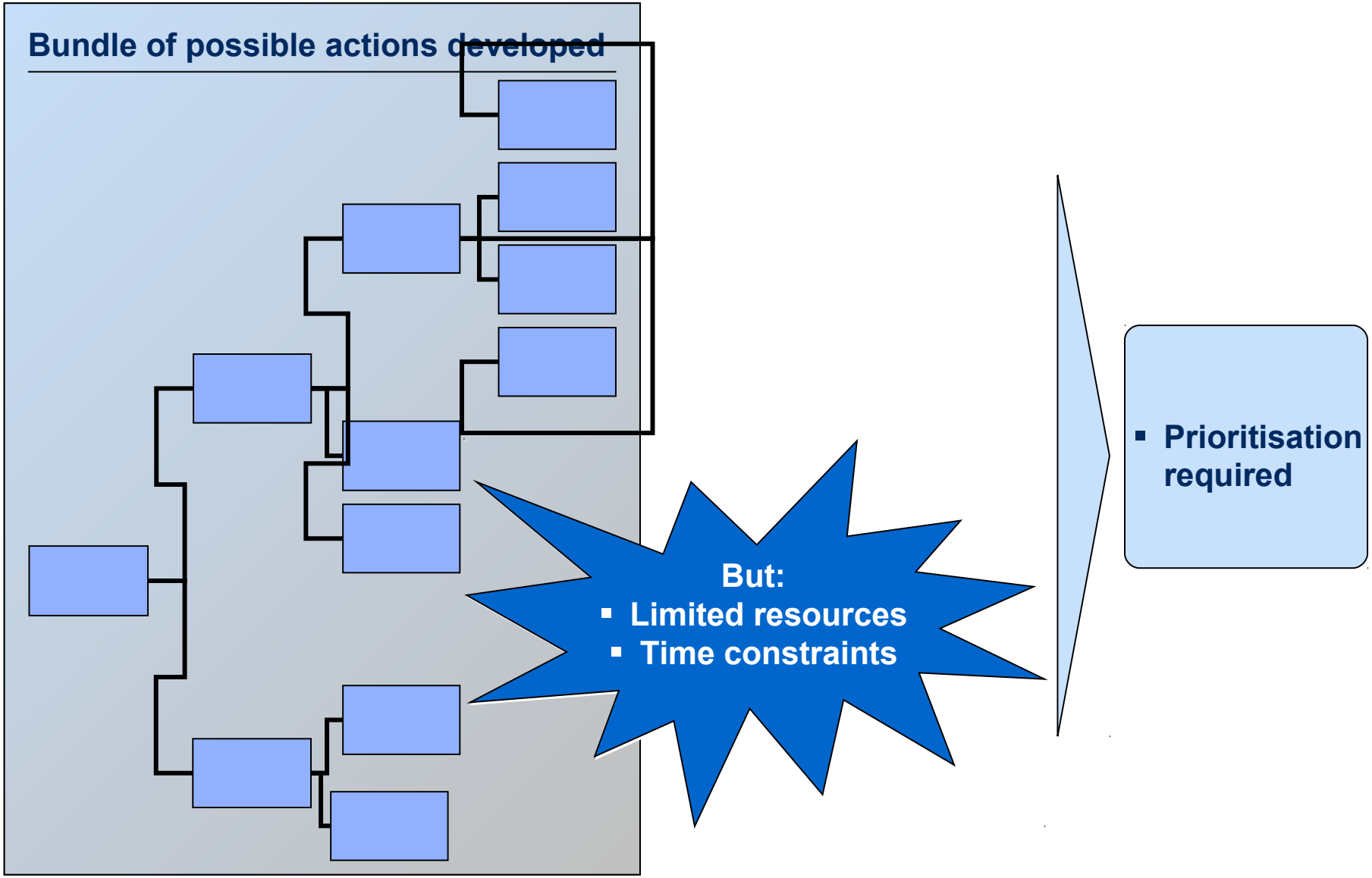


Good logic trees help prioritise issues and allocate responsibilities to team members

Think **SPEED**:
Which part of the tree seems most important to the problem?



Purpose of prioritisation

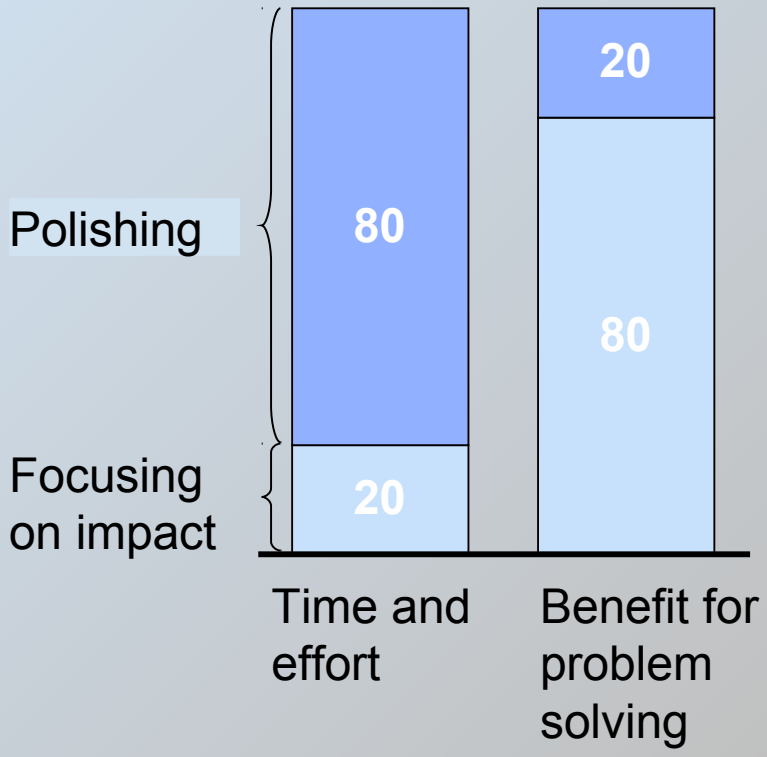


How to prioritise

Use judgement/intuition

Involve your team

Be practical!



Do back-of-the-envelope calculations

Take risks

Methods for prioritising

Key ideas

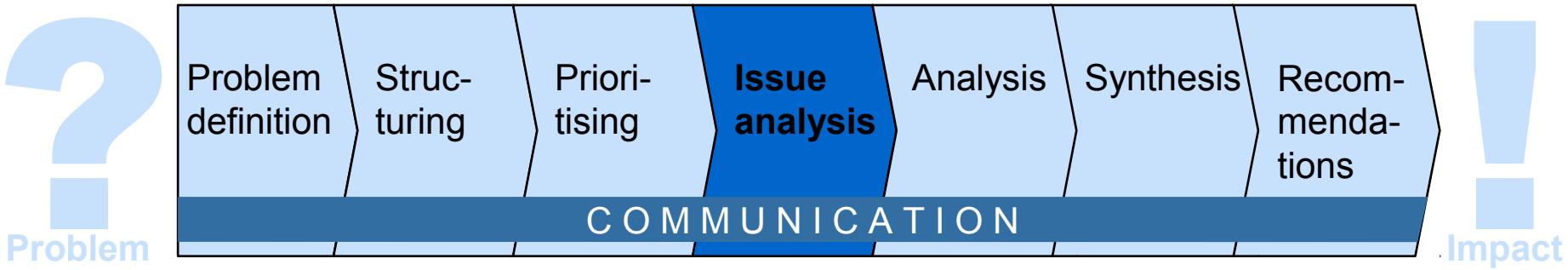
- Take a reasonably quick, informal approach to get started
- Remember that sometimes a ”back of an envelope” calculation is all that’s needed
- If new information emerges, you can always reprioritise and switch your efforts to another part of the tree
- Use likely impact to decide where to go first
- Use readily available data whenever possible; avoid major data requests
- Do not create massive spreadsheets or other computer models if you can move forward with less complete information

Example methods

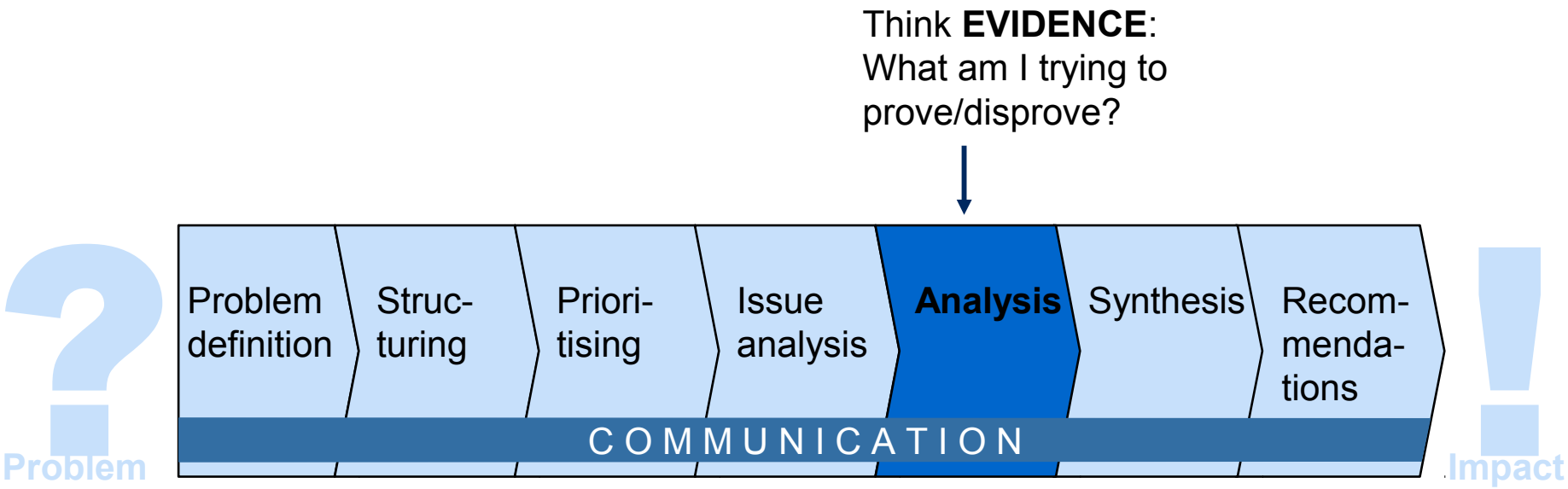
- Percentage of total “X”
- Estimated potential increase/decrease (sensitivity analysis)
- Simple ratios
- Quick industry benchmarks
- Qualitative input from interviews
- Ease of implementation
- Timing
- “Quick wins”
- Available resources
- High/low risk

Step 4: Issue analysis and work plan

Think **EFFICIENCY**:
How and on what should the team spend its time?

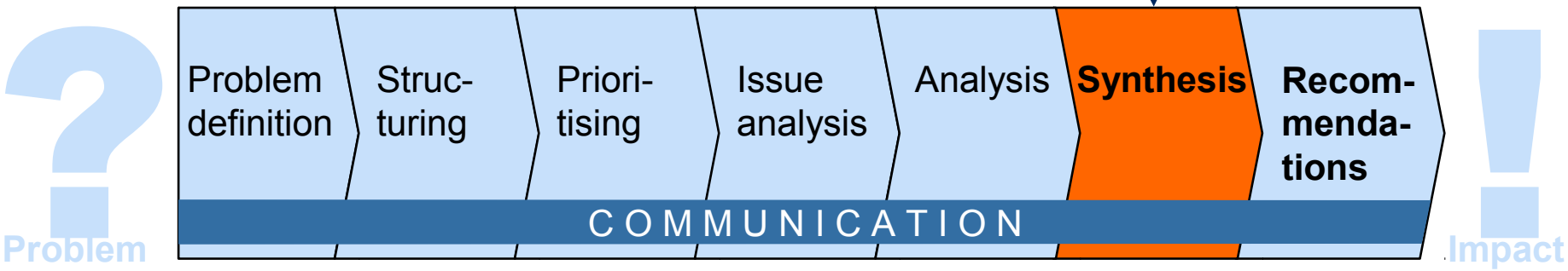


Step 5: Analysis



Step 6: Synthesise findings

Think **SO WHAT**:
What are the implications of our findings?



Summary versus synthesis: what's the difference?

SUMMARY

?

SYNTHESIS

?

FACTS

- I have misplaced my keys
- My passport isn't where I thought it was
- I'm 2 months behind on my tax return

Summary versus synthesis: extracting a higher level of meaning

SUMMARY

I've lost my keys and passport and I'm behind on my tax return

SYNTHESIS

I've been sloppy

FACTS

- I have misplaced my keys
- My passport isn't where I thought it was
- I'm 2 months behind on my tax return

Synthesis helps develop a powerful and effective main message

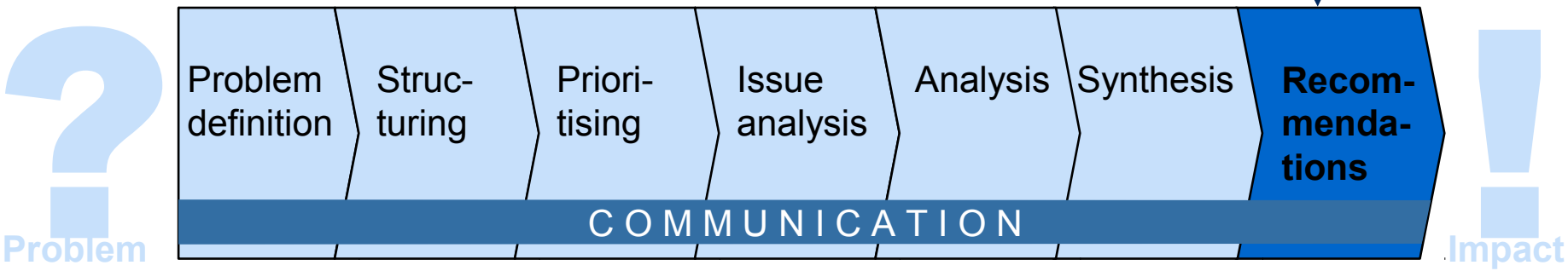
What is the one thing I want my audience to think or do as a result of this communication?

Main message must be:

- **T**argeted
- **O**verarching
- **P**owerful
- **S**upportable

Step 7: Develop recommendations

Think **POTENTIAL SOLUTION**: What should be done?



Last chance for any questions…

Thank you – Enjoy the rest of your day